

No. of Functional Features Included in the Solution

Date	7 July 2026
Team ID	XXXXXX
Project Name	OptiCrop – Crop Recommendation System
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Input Form	Allows users to enter Nitrogen, Phosphorus, Potassium, Temperature, Humidity, pH, and Rainfall values.	User Interface	Done	High
2	Input Validation	Validates numerical input and prevents invalid data submission.	Backend	Done	High
3	Crop Prediction	Predicts the most suitable crop using the trained machine learning model.	ML Prediction Engine	Done	High
4	Crop Description	Displays a brief description of the recommended crop.	Backend	Done	Medium
5	Responsive Web Interface	Provides a simple and responsive interface for desktop and mobile users.	Frontend	Done	Medium
6	Model Loading	Loads the trained crop_model.pkl file using Joblib/Pickle.	Backend	Done	High
7	Deployment on Render	Hosts the application online for public access.	Deployment	Done	High
8	Error Handling	Handles invalid inputs and prediction errors gracefully.	Backend	Done	Medium

Feature Summary:

Metric	Count / Value
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Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Input Form, Responsive Design
2	Backend / Logic	4	Input Validation, Crop Prediction, Crop Description, Error Handling
3	Database / Storage	1	Trained Model (crop_model.pkl) and CSV Dataset (File Storage)
4	API / Integration	1	Flask Integration with Machine Learning Model
5	Security / Authentication	0	Not Implemented (No Login Required)